



Assignment 2 — Build It, Then Break It

(Lecture: Context Engineering)

Goal: Design a layered system prompt for a production assistant and adversarially red-team it.

Task

Scenario (or propose your own of similar complexity): *A customer support assistant for a mid-size SaaS company. It answers product questions, checks order status via a (pretend) tool, and can issue refunds up to \$50 without approval. It must never discuss competitors, never promise unshipped features, and must escalate legal threats or abuse.*

- **Write a layered system prompt** with clearly labeled blocks: identity/persona → scope & capabilities → hard constraints → business-policy constraints → style → output format. Mark which layer each rule belongs to.
- **Red-team it** against all 10 attack categories from the lecture (direct injection, role-play jailbreak, prompt extraction, scope creep, fabricated authority, urgency/social engineering, data exfiltration, emotional pressure, obfuscation, multi-turn erosion). For multi-turn erosion, run at least a 6-turn conversation.
- **Log every break** in a break log: attack → what the model did → severity (leaked info / violated policy / just went off-tone).
- **Ship v2:** patch the prompt, re-run the attacks that broke v1, and report which fixes worked and which didn't.
- **Draw the line:** pick 2 constraints from your prompt and argue whether each should be enforced by the prompt or by a hard-coded guardrail outside the model — and why.

Deliverables

- v1 and v2 system prompts (with layer labels)
- Break log covering all 10 categories (transcripts or excerpts)
- v1→v2 diff summary: what changed and why
- The 2-constraint guardrail analysis (½ page)





Grading (100 pts)

Criterion	Pts
Prompt is genuinely layered with correct constraint hierarchy	25
All 10 attack categories tested honestly (including a real multi-turn erosion attempt)	30
Break log is specific: what broke, how badly, evidence	20
v2 fixes are targeted at logged breaks; re-test shown	15
Guardrail-vs-prompt analysis shows real judgment	10

Note: attacks stay inside your own sandbox against your own prompt. Do not test on live products you don't own.

Submission

Submit a single PDF or repo. Include model name/version and temperature for every run. Raw outputs must be unedited — cherry-picking a lucky run counts against the honesty criteria.

